Simpson's Rule

Another area approximating tool is Simpson's Rule.
Geometrically, it creates tiny parabolas to wrap closer around the function we are approximating.
The formula is similar to the Trapezoidal Rule, with a small catch...
you can only use an even number of subintervals.

Definition

Let *f* be continuous on [*a*, *b*].
Simpson's Rule for approximating $\int_a^b f(x)\,dx$ is given by

$$\frac{b-a}{3n}[f(a)+4f(x_1)+2f(x_2)+\cdots+4f(x_{n-1})+f(b)]$$

Simpson's Rule A Second Glimpse

$$\int_a^b f(x)\,dx=\frac{h}{3}[y_0+4y_1+2y_2+4y_3+\cdots+2y_{n-2}+4y_{n-1}+y_n]$$

where [*a*, *b*] is partitioned into *n* even subintervals of equal length.

$$\Delta x=h=\frac{b-a}{n}$$

Ex. 1: Approximate the area beneath $f(x)=x^2+1$ on the interval $[0,3]$ using Simpson's Rule with $n=6$ subintervals.

Solution:

$$\Delta x=h=\frac{b-a}{n}=\frac{3-0}{6}=\frac{1}{2}$$

x	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3
$f(x)=y$	1	$\frac{5}{4}$	2	$\frac{13}{4}$	5	$\frac{29}{4}$	10

$$I=\frac{h}{3}[y_0+4y_1+2y_2+4y_3+2y_4+4y_5+2y_6]$$

$$I=\frac{1}{2(3)}\left[1+4\left(\frac{5}{4}\right)+2(2)+4\left(\frac{13}{4}\right)+2(5)+4\left(\frac{29}{4}\right)+10\right]$$

$$I=\frac{1}{6}[1+5+4+13+10+29+10]$$

$$I=\frac{1}{6}[72]=12$$

Ex. 2: Approximate the area beneath $y = sinx$ on the interval $[0, \pi]$ using the Simpson's Rule with $n = 6$ subintervals.

Solution:

$$\Delta x = h = \frac{b-a}{n} = \frac{\pi - 0}{6} = \frac{\pi}{6}$$

x	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π
$f(x) = y$	0	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	0

$$I = \frac{h}{3}[y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + 4y_5 + y_6]$$

$$I = \frac{\pi}{6(3)}\left[0 + 4\left(\frac{1}{2}\right) + 2\left(\frac{\sqrt{3}}{2}\right) + 4(1) + 2\left(\frac{\sqrt{3}}{2}\right) + 4\left(\frac{1}{2}\right) + 0\right]$$

$$I = \frac{\pi}{18}\left[0 + 2 + \sqrt{3} + 4 + \sqrt{3} + 2 + 0\right]$$

$$I = \frac{\pi}{18}\left[8 + 2\sqrt{3}\right]$$

$$I \approx 2.001$$

Ex. 3: Find approximate value of the following integrations by Simpson's rule $I = \int_0^4 x\, ln\, x\, dx$ where $n = 8, a = 0, b = 4$.

$$h = \frac{b-a}{n} = \frac{4-0}{8} = 0.5$$

$f(x) = x\, ln\, x$

x_i	0	0.5	1	1.5	2	2.5	3	3.5	4
$f(x_i) = y_i$	0	-0.347	0	0.608	1.386	2.291	3.296	4.385	5.545

$$I = \frac{h}{3}(y_0 + 4(y_1 + y_3 + y_5 + y_7) + 2(y_2 + y_4 + y_6) + y_8)$$

$$I = \frac{0.5}{3}(0 + 4(-0.347 + 0.608 + 2.291 + 4.385) + 2(0 + 1.386 + 3.296) + 5.545)$$

$$= 0.167(0 + 4(6.937) + 2(4.682) + 5.545) = 0.167(27.748 + 9.364 + 5.545)$$

$$= 0.167(42.657) = 7.124$$

Ex. 4: Use the Simpson's rule of integration to determinate the value of the following integrals $I = \int_1^2 \frac{\cos x}{x} dx$, where $n = 8, a = 1, b = 2$.

$$h = \frac{b-a}{n} = \frac{2-1}{8} = 0.125$$

$f(x) = \frac{cos\, x}{x}$

x_i	1	1.125	1.25	1.375	1.5	1.625	1.75	1.875	2
$f(x_i) = y_i$	0.5403	0.3833	0.2523	0.1415	0.0472	-0.0333	-0.1019	-0.1598	-0.2081

$$I = \frac{h}{3}\left(y_0 + y_8 + 4(y_1 + y_3 + y_5 + y_7) + 2(y_2 + y_4 + y_6)\right)$$

$$I = \frac{0.125}{3}(0.5403 - 0.2081 + 4(0.3833 + 0.1415 - 0.0333 - 0.1598) + 2(0.2523 + 0.0472 - 0.1019)) = 0.0417(0.3322 + 4(0.3317) + 2(0.1976)) = 0.0417(2.0542) = 0.0856$$